



COLUMBIA UNIVERSITY

Department of Statistics

Completed in Partial Fulfillment of the Requirements of

STAT S4204

Statistical Inference

Summer 2026

ASSIGNMENT 2

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Contents

Probability Review (Problems 1–7)	1
Problem 1: Good Faith and Retrieval	1
Problem 2: Expectation	2
Problem 3: Variance	4
Problem 4: Conditional Distribution	6
Problem 5: Conditional Expectation	8
Problem 6: Conditional Expectation as a Random Variable	10
Problem 7: Iterated Expectations	12
The Decision-Theoretic Vocabulary (Problems 8–14)	14
Problem 8: The Statistical Model	14
Problem 9: The Analytic Goal	16
Problem 10: The Three Elements of a Decision Problem	18
Problem 11: Two Definitions: Risk and Admissibility	20
Problem 12: The Bias–Variance Decomposition	22
Problem 13: Admissibility	24
Problem 14: Comparing Decision Rules	26
Risk Computations (Problems 15–16)	28
Problem 15: Two Risk Functions for the Binomial	28
Problem 16: The Risk Ratio in the Limit	31
References	32

Probability Review (Problems 1–7)

Problem 1

Repeat to yourself: these problem sets are graded on good faith effort; learning is effected on a cellular level when you retrieve dimly remembered concepts (and not when you look them up with AI); it may take longer to try, but at least it is not wasted time.

SOLUTION:

Repeated—and adopted as the working protocol for these write-ups. Everything below was drafted from memory first and checked against the texts and the lecture notes afterwards; where the from-memory draft proved wrong or incomplete, the corrected version is what appears, which is exactly the retrieve-then-verify loop the instruction prescribes. The same advice closes Lecture 1 itself (notes, Remark 1.8.6): set up the calculations yourself, because “that is practicing the conceptual issues, and that is the point.” Problems 15 and 16 are worked accordingly. ■

Problem 2

Probability review: explain the concept of expectation.

SOLUTION:

The expectation of a random variable is the probability-weighted average of its values—the single number that locates the center of its distribution. For discrete Y with pmf p and continuous Y with density f ,

$$\mathbb{E}\{Y\} = \sum_y y p(y) \quad \text{or} \quad \mathbb{E}\{Y\} = \int y f(y) dy,$$

provided the sum or integral converges absolutely. Two features of this definition do real work. First, expectations belong to *distributions*: a random variable has an expectation only through its law (notes, Def. 1.4.1), fixed before any data are seen—which is what separates the expectation from a sample average, an arithmetic operation on realized numbers that merely *estimates* it. Second, the absolute-convergence proviso is not decoration: the values of a distribution come in no privileged order, and a conditionally convergent sum can be rearranged to any total, so without it “the” expectation would depend on bookkeeping. DeGroot and Schervish build the caveat into a four-part definition (Defs. 4.1.1–4.1.4, pp. 208–210), with the Cauchy distribution as the standard casualty: no mean at all.

The interpretations are as standard as the formula. It is a center of mass—place the probability as weight along the line and $\mathbb{E}\{Y\}$ is the balancing point, “analogous to the physical concept of the center of gravity” (Ross, p. 121; the seesaw picture is Blitzstein and Hwang’s Figure 4.1, p. 150). It is a long-run average, the value the sample mean approaches over many replications (Ross, p. 119; Pitman’s boxed version, p. 164). And

it is a fair price: “ $E(X)$ is the fair price to pay for a return of X ” (Pitman, p. 166), which is how an insurer prices a policy—the premium starts from the expected payout. A fair die makes all three readings concrete at once: $E\{Y\} = 3.5$, the balance point of six equal masses, the long-run average of many rolls, a value the die itself never shows. That last detail is worth keeping: an expectation is a property of the distribution, not necessarily a possible outcome. ■

Problem 3

Probability review: explain the concept of variance.

SOLUTION:

Where the expectation locates a distribution, the variance measures how badly a draw typically misses that location—the expected squared deviation. For Y with $\mu = E\{Y\}$ and $E\{Y^2\}$ finite,

$$\text{Var}\{Y\} = E\{(Y - \mu)^2\} = E\{Y^2\} - (E\{Y\})^2,$$

the shortcut form by expanding the square and applying linearity. Variance is non-negative, is zero exactly when Y is constant with probability one, and carries squared units; its square root, the standard deviation, lives on the scale of Y itself, and Pitman is candid about why the pair persists: “Variance is harder to interpret than SD, but has simpler algebraic properties” (p. 185). It is not linear—constants drop out and scale factors emerge squared, $\text{Var}\{aY + c\} = a^2 \text{Var}\{Y\}$ —a rule Problem 15 will lean on twice.

What this course adds to the standard account is a change of role. In the probability texts, variance is descriptive spread, motivated by the algebraic inconvenience of the absolute deviation (Ross, p. 126; Blitzstein and Hwang, pp. 171–172). In the decision-theoretic frame of Lecture 1 it becomes a performance measure: under squared-error loss the risk of an estimator is its mean squared error, and “the mean squared error $E(d - \theta)^2$ becomes the variance of d when $E(d) = \theta$ ” (HPS, p. 15)—for unbiased estimators, variance *is* the risk. The two thermometers of everyday experience carry the idea: both read the true temperature on average, but the one that scatters $\pm 2^\circ$

is a worse instrument than the one that scatters $\pm 0.2^\circ$, and variance is precisely the number that says so. Problem 12 extends this reading to biased estimators, and Problem 15 computes it in a live comparison. ■

Problem 4

Explain the concept of conditional distribution.

SOLUTION:

The conditional distribution of Y given $X = x$ is the distribution of Y for someone who has learned that $X = x$: it is how a distribution absorbs information. For any x with $p_X(x) > 0$,

$$p_{Y|X}(y | x) = \frac{p(x, y)}{p_X(x)},$$

and $f_{Y|X}(y | x) = f(x, y)/f_X(x)$ in the density case. The recipe behind the formula is slice and renormalize: fix $X = x$, take that slice of the joint distribution—the x -row of the two-way table—and rescale it by its total so it sums to one (notes, Remark 1.4.11). Nothing here is new probability: since $\{Y = y\} \cap \{X = x\}$ has probability $p(x, y)$, the display is the elementary $P(A | B) = P(A \cap B)/P(B)$ applied to every event $\{Y = y\}$ at once, which is what upgrades a single conditional probability into an entire conditional distribution. Ross introduces it exactly this way, “natural to define” by analogy with $P(E | F)$ (§6.4, p. 248).

The continuous case hides a genuine subtlety that the texts handle with different degrees of candor. The conditioning event $\{X = x\}$ has probability zero, so the density formula cannot be *derived* from the elementary definition; Ross notes the difficulty and proceeds on the strength of the infinitesimal reading (p. 251), Pitman makes that reading the official one with his infinitesimal conditioning formula $P(A | X = x) = P(A, X \in dx)/P(X \in dx)$ (p. 410, with the vertical-slice picture on p. 412), and DeGroot and Schervish are the most explicit that the formula is a *definition*, not a theorem—where $f_X(x) = 0$ “we are free to define” the conditional density however we

wish (Def. 3.6.2, p. 144). The fully rigorous resolution, for reference, is the theory of regular conditional distributions, well beyond this course. One consequence deserves its own sentence, because the next problem is built on it: for each fixed x , the conditional distribution is a genuine probability distribution—nonnegative, summing to one—so everything defined for distributions applies to it verbatim (notes, Remark 1.4.14). A concrete instance: the distribution of a commute time given that it is raining is a different distribution from the marginal one—shifted right, probably widened—and the conditional distribution is exactly the object that says how. ■

Problem 5

Explain the concept of conditional expectation.

SOLUTION:

Because a conditional distribution is a genuine probability distribution (Problem 4), it has an expectation, and that is the whole definition: the conditional expectation of Y given $X = x$ is the expectation *of the conditional distribution*,

$$E\{Y \mid X = x\} = \sum_y y p_{Y|X}(y \mid x)$$

(with $\int y f_{Y|X}(y \mid x) dy$ in the density case): the best single-number summary of Y for someone who has learned that $X = x$ (notes, Def. 1.4.15; Ross, §7.5.1, pp. 313–314; DeGroot and Schervish, Def. 4.7.1, p. 256; Pitman, p. 401, as the expectation “under the conditional probability distribution”). Ross’s gloss is the right mental model: an ordinary expectation “on a reduced sample space” (p. 315)—condition, then average, with no new machinery.

As x varies, these numbers trace out a function of x —the regression function, the object at the center of the linear-model weeks ahead. The worked table in the notes shows the whole apparatus in miniature (Example 1.4.12): there $E\{Y \mid X = 0\} = 1.8$ and $E\{Y \mid X = 1\} = 2.3$, two slice-averages summarizing two conditional distributions; learning $X = 1$ tilts the prediction of Y upward. And “best” is not a figure of speech: among all functions of X , the conditional expectation minimizes the mean squared prediction error $E\{(Y - g(X))^2\}$ —the projection property (Blitzstein and Hwang, Thm. 9.3.9, p. 429; DeGroot and Schervish, Thm. 4.7.3, p. 260; Ross, Prop. 6.1, p. 330; Pitman files it as Exercise 6.2.17, p. 409). Under the squared-error loss this course has

adopted, conditional expectation is therefore not one summary among many; it is the optimal prediction of Y from X , which is why it will keep reappearing. ■

Problem 6

What is the difference between $E\{Y \mid X\}$ and $E\{Y \mid X = x\}$?

SOLUTION:

$E\{Y \mid X = x\}$ is a *number*; $E\{Y \mid X\}$ is a *random variable*. The number is the expectation of the conditional distribution of Y given $X = x$, computed for one fixed x ; collecting these numbers over all x defines a function, and evaluating that function at the random X —before it crystallizes into a realization—produces the random variable:

$$E\{Y \mid X\} = g(X), \quad \text{where} \quad g(x) = E\{Y \mid X = x\}$$

(notes, Def. 1.4.15). Every text on the shelf makes precisely this two-step move, in almost the same words. Ross: “Let us denote by $E[X \mid Y]$ that function of the random variable Y whose value at $Y = y$ is $E[X \mid Y = y]$... Note that $E[X \mid Y]$ is itself a random variable” (§7.5.2, p. 315). DeGroot and Schervish: let $h(x)$ stand for $E(Y \mid x)$; “define the symbol $E(Y \mid X)$ to mean $h(X)$ ” (Def. 4.7.2, p. 257). Pitman boxes it: “the function of X whose value is $E(Y \mid X = x)$ if $X = x$,” adding that it “is actually a random variable” (pp. 402–403). Blitzstein and Hwang give the definition (Def. 9.2.1, p. 425) and then post the warning this problem is really about: quantities of the form $E(Y \mid A)$ “are numbers, while those of the form $E(Y \mid X)$ are random variables”—confusing the two is a category error (their hazard boxes 9.2.2–9.2.3, p. 425).

Figure 1 draws the distinction. The dot on the slice at x_0 is the number $E\{Y \mid X = x_0\}$: one conditional distribution, averaged. The line through the contours is the graph of g ; the random variable $E\{Y \mid X\}$ is that line evaluated wherever X happens to land. The payoff of the random-variable view is that $E\{Y \mid X\}$ has a

distribution of its own, hence an expectation—which is exactly Problem 7’s rule—and a variance, which splits $\text{Var}\{Y\}$ into within-slice and between-slice parts (the conditional-variance decomposition of DeGroot and Schervish, Thm. 4.7.4, p. 261, and Blitzstein and Hwang, Thm. 9.5.5, p. 434).

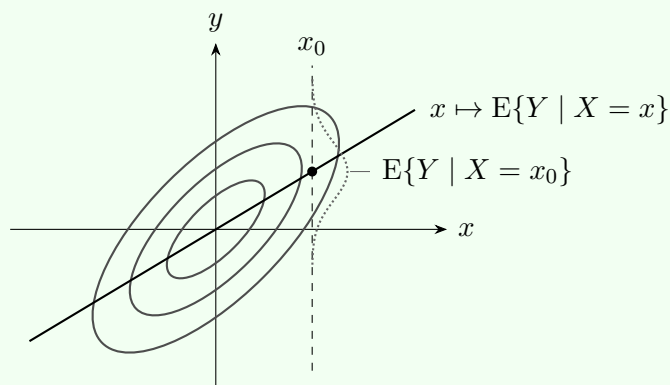


Figure 1. A NUMBER AND A RANDOM VARIABLE. Contours of a joint density, with the conditional density of Y along the slice $X = x_0$ sketched sideways (dotted). Averaging that slice gives the number $E\{Y \mid X = x_0\}$, the marked point. Doing so for every x traces the solid line, the function g ; the random variable $E\{Y \mid X\} = g(X)$ is that line evaluated at wherever X lands.



Problem 7

What is the rule for iterated expectations?

SOLUTION:

Whenever the expectations exist,

$$\boxed{E\{E\{Y \mid X\}\} = E\{Y\}}$$

(notes, Thm. 1.4.16). In words: to average Y , first average it within each slice $X = x$, then average the slice-averages, weighting each by how likely its slice is. The left side is well defined only because Problem 6 made $E\{Y \mid X\}$ a random variable; the proof is then three moves—write the outer expectation by the law of the unconscious statistician, $\sum_x E\{Y \mid X = x\} p_X(x)$; substitute the conditional expectation and cancel $p_X(x)$ against the conditional pmf’s denominator, leaving $\sum_x \sum_y y p(x, y)$; exchange the order of summation (absolute convergence again earning its keep) and recognize the marginal of Y (the notes work both the discrete and density versions, Proofs 1.4.16.a–b). The worked table of Example 1.4.12 shows it numerically: the slice-averages 1.8 and 2.3, weighted by $p_X = (0.5, 0.5)$, give 2.05, which is $E\{Y\}$ computed directly.

The rule is standard everywhere and named nowhere the same way. Ross states it as Proposition 5.1 under the rubric “computing expectations by conditioning” (§7.5.2, p. 315); DeGroot and Schervish call it the “Law of Total Probability for Expectations” (Thm. 4.7.1, p. 258); Pitman boxes it as “Expectation is the Expectation of the Conditional Expectation” (p. 403), the condensed form of his rule of average conditional expectations (p. 402); Blitzstein and Hwang christen it Adam’s law (Thm. 9.3.7, p. 427)—their stated reason being that a rule “used so frequently... deserves a pithy

name”—pairing it with Eve’s law for the variance. For this course its work is ahead of it: risks are expectations of losses, Bayes risks are expectations of risks, and both will routinely be computed exactly this way, by conditioning on whatever makes the inner average easy. ■

The Decision-Theoretic Vocabulary (Problems 8–14)

Problem 8

What do we mean (in this course, at least) by a statistical model, mathematically, and “in the real world?”

SOLUTION:

Mathematically: *the statistical model is the set of all possible distributions of the data* (notes, Def. 1.3.1). If the data are random they have a joint distribution, and—in the lecture’s own emphasis—all you can learn about is the data and the distribution of the data, “and that is all.” If the study has a goal, something about that distribution must be unknown, so the model must contain more than one candidate. In practice the family is indexed: a parameterization assigns to each θ in a parameter space Θ one distribution, and the model is written

$$\{p^\theta \text{ or } f^\theta : \theta \in \Theta\}$$

with p^θ a pmf or f^θ a density for the data (Def. 1.3.2). The clinical trial makes it concrete: two arms of independent Bernoulli outcomes, $\theta = (\theta_T, \theta_P) \in [0, 1]^2$, and $p^\theta(y) = \theta_T^{x_T} (1 - \theta_T)^{n_T - x_T} \theta_P^{x_P} (1 - \theta_P)^{n_P - x_P}$ (Example 1.8.4).

In the real world, the model is a record of knowledge and ignorance in exactly the proportions you actually possess them. What you *know* about the phenomenon and the data collection determines which distributions are plausible—independence across patients, a common cure rate within an arm; what you *don’t know* is why the model contains more than one distribution at all (notes, Concept 1.3.5, and the pipeline of

§1.2). The course text never says “statistical model”: HPS speaks of a *probability model* “appropriate for representing the occurrence of the phenomenon,” with the distribution of X “assumed to belong to a certain family of distributions” (p. 1)—the same object, family-first. DeGroot and Schervish give the most expansive version on the shelf: their statistical model identifies the random variables of interest (“both observable and only hypothetically observable”), a family of joint distributions, the parameters, and, if desired, a prior for them (Def. 7.1.1, p. 377)—a definition already built to house the Bayesian material this course reaches in §1.6 of the notes. ■

Problem 9

What do we mean (again, in this course) by the analytic goal?

SOLUTION:

The analytic goal is an expression of the purpose of the study *in terms of the parameters of the model*: the scientific question, which lives in the real world, translated into a statement about θ (notes, Def. 1.3.3). Schematically,

$$\boxed{\text{scientific question} \longmapsto \text{statement about } \theta}.$$

The translation is the point. “Is the treatment efficacious?” is a question about reality; it is not yet something mathematics can act on. Once the trial’s model is fixed—two Bernoulli arms with parameters (θ_T, θ_P) —the question becomes *learn the difference* $\theta_T - \theta_P$ (Example 1.8.4), and that is an object the machinery of Problems 10–14 can process: methods can aim at it, losses can price errors about it, risks can be computed for it. In the pipeline of the lecture’s §1.2, the analytic goal is the box fed jointly by the statistical model and the goal of the study: the model supplies the vocabulary of parameters, the study supplies the purpose, and the analytic goal is the purpose rewritten in that vocabulary.

The term itself is the course’s own; the texts have only near relatives. HPS frames the counterpart as “the problem of statistical inference”—deciding “which member, or members, of the family could represent the distribution of X ” (p. 1)—and DeGroot and Schervish define a statistical inference as “a procedure that produces a probabilistic statement about some or all parts of a statistical model” (Def. 7.1.2, p. 378). Both describe the activity; the deck’s phrase names its target, and keeping the name separate

from the machinery is what lets the same trial support estimation one day and testing the next (Problem 10): the analytic goal stays fixed while the action space changes. ■

Problem 10

For the elements of decision problem, we said that there were three. What were they? (This is before we get to Bayesian decision theory.)

SOLUTION:

Decision theory begins from the statistical model of Problem 8 and adds three objects (notes, §1.5):

$$\mathcal{A} \text{ (action space),} \quad L : \mathcal{A} \times \Theta \rightarrow \mathbb{R} \text{ (loss function),} \quad \delta : S \rightarrow \mathcal{A} \text{ (method)}.$$

The *action space* $\mathcal{A}$ is the set of possible choices of the decision maker—the values an analysis is allowed to output (Def. 1.5.1). It is fixed by the question, not by the data: the same trial supports estimation, $\mathcal{A} = [-1, 1]$, every value a difference of two proportions could take, or testing, an action space with exactly two elements—and changing $\mathcal{A}$ changes the problem. The *loss function* prices consequences: $L(a, \theta)$ is how much you lose by choosing action a when the truth is θ (Def. 1.5.3)—a modeling choice as much as the model itself, declaring in advance how bad each mistake would be. This course will keep to squared-error loss for estimation and zero-one loss for testing (Defs. 1.8.1–1.8.2). The *method*, or decision rule, is a function δ from the sample space to the action space: what you will do, specified in advance, for every data set you might observe (Def. 1.5.5)—a complete contingency plan, not an answer, and that timing is what makes methods judgeable, since a plan can be averaged over everything the model says the data may do.

The count is worth defending, because the assignment’s parenthesis explains it. Bayesian decision theory (notes, §1.6) adds exactly one further object—the prior dis-

tribution on Θ —so *before* the Bayesian turn, the inventory is: model, plus these three. The vocabulary is Wald’s, and the course text already carries it in 1971 dress: HPS has the decision function $d(x_1, \dots, x_n)$ (p. 3), actions a_1, a_2 (p. 3), and the loss $\mathcal{L}(\theta, a)$ (p. 5)—with the amusing wrinkle that the argument order varies by source: HPS and this assignment write the loss with θ first, the deck writes $L(a, \theta)$, and nothing depends on the choice (for squared error, $(p - a)^2 = (a - p)^2$). Of the probability texts only DeGroot and Schervish define a loss function at all (Def. 7.4.2, p. 409); the machinery is native to the statistics shelf. ■

Problem 11

We also had a couple of definitions. What were they?

SOLUTION:

After the three elements, the lecture laid down exactly two definitions: the *risk function* and *admissibility* (notes, Defs. 1.5.9 and 1.5.13). The risk function of a decision rule δ is its expected loss, as a function of the parameter:

$$\boxed{R_\delta(\theta) = E^\theta\{L(\delta(Y), \theta)\}} = \sum_{y \in S} p^\theta(y) L(\delta(y), \theta) \quad \text{or} \quad \int_{y \in S} f^\theta(y) L(\delta(y), \theta) dy,$$

where E^θ is expectation under the distribution indexed by θ . Read it from the inside out: fix a candidate truth; the model hands the data their distribution; the method turns random data into a random action; the loss turns the action into a random penalty; and the expectation—the display is the law of the unconscious statistician with $g(y) = L(\delta(y), \theta)$ —collapses the randomness to one number, the method’s average penalty *if θ were the truth*. Sweeping θ across Θ produces not a number but a curve, a point the lecture insists on: “Remember, Risk is a function!” The second definition, admissibility, is the discipline that curves make necessary—a method is admissible when no other method matches it everywhere and beats it somewhere; its full statement is Problem 13.

Two shelf notes. In HPS, the risk function is literally *Definition 1 of the book* ($\mathcal{R}(\theta, d) = E_\theta \mathcal{L}(\theta, d(X_1, \dots, X_n))$, p. 6)—a statement about where that text believes statistics begins. And the one place the probability shelf brushes against any of this is a single page of Pitman: the expected loss “or risk” $r(b) = E[L(X, b)]$ (p. 178), with a figure captioned “Risk functions for a die roll” (p. 179) showing squared error mini-

mized by the mean, absolute error by the median, zero–one loss by the mode—decision theory in miniature, three losses electing three different summaries. ■

Problem 12

Probability review: what is the “bias plus variance decomposition” of the “mean squared error?”

SOLUTION:

Let $\delta = \delta(Y)$ estimate a target $g(\theta)$ under squared-error loss, and write $m(\theta) = E^\theta\{\delta\}$ for the estimator’s mean and $b(\theta) = m(\theta) - g(\theta)$ for its *bias*. Then the mean squared error splits exactly in two:

$$E^\theta\{(\delta - g(\theta))^2\} = \text{Var}^\theta(\delta) + b(\theta)^2.$$

The proof is one add-and-subtract: insert $m(\theta)$,

$$E^\theta\{(\delta - m(\theta) + m(\theta) - g(\theta))^2\} = E^\theta\{(\delta - m(\theta))^2\} + 2b(\theta)E^\theta\{\delta - m(\theta)\} + b(\theta)^2,$$

and the cross term dies because $E^\theta\{\delta - m(\theta)\} = 0$; what remains is variance plus squared bias (notes, Supplement 1.A.1). The reading matters as much as the identity: under squared-error loss there are exactly two ways to be wrong—scatter around your own average (variance) and a systematic offset of that average from the target (bias)—and risk charges for both, in the same currency. Figure 2 draws the anatomy. An unbiased estimator sets $b \equiv 0$, and the decomposition collapses to the sentence HPS builds §2.1 on: the mean squared error “becomes the variance of d when $E(d) = \theta$ ” (p. 15). The general identity, curiously, is not in the course text at all—nor in Ross or Pitman; Blitzstein and Hwang assign it as an exercise (Ch. 5, Exercise 53, p. 263, “a form of the bias-variance tradeoff”), and among the five books only DeGroot and Schervish state it outright: the M.S.E. of an estimator “equals its variance plus the

square of its bias” (Cor. 8.7.1, p. 507).

The decomposition is the engine of every risk computation in this assignment: it reduces Problem 15 to a mean, a variance, and one subtraction. And it frames the course’s first genuine trade-off—a little bias can be worth buying if it comes with a large enough variance reduction, which is exactly the wager the shrunk estimator d_1 of Problem 15 makes.

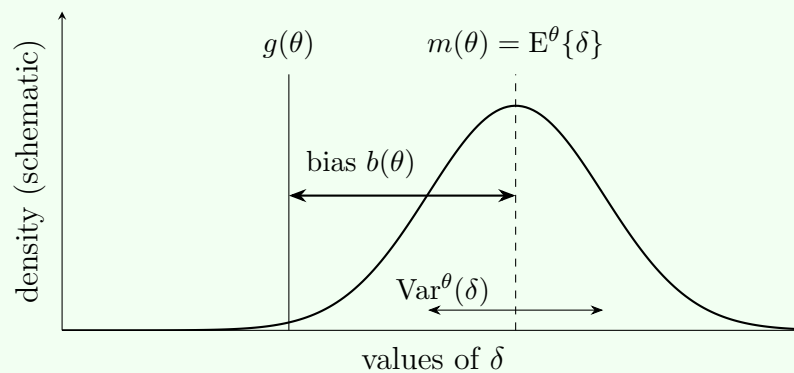


Figure 2. THE TWO WAYS TO MISS. The sampling density of an estimator δ under a fixed θ : its draws scatter about their own mean $m(\theta)$ (variance), and that mean sits at a systematic offset $b(\theta)$ from the target $g(\theta)$ (bias). Squared-error risk is the sum $\text{Var}^\theta(\delta) + b(\theta)^2$: both defects, one currency.



Problem 13

What is the definition of admissibility?

SOLUTION:

A method is *admissible* if there is no other method that does at least as well for every parameter value and better for some parameter value (notes, Def. 1.5.13). Formally,

$$\delta \text{ admissible} \iff \text{no } \delta' \text{ has } R_{\delta'}(\theta) \leq R_{\delta}(\theta) \text{ for all } \theta \text{ and } < \text{ for some } \theta .$$

A method that is so dominated is *inadmissible*, and the lecture’s gloss explains why the word carries normative weight: whatever θ turns out to be, you would not regret abandoning it. Two features of the definition deserve emphasis. It is *pairwise and pointwise*—domination must hold at every θ simultaneously, which is a demanding standard, so risk curves that cross (as in Problem 15, Figure 3) leave both methods admissible. And it only *prunes*: admissibility discards the dominated and then falls silent, leaving a family of survivors among which it cannot choose; picking one is the business of Bayesian decision theory, which the assignment has fenced off.

The concept’s location in the literature is itself instructive. HPS’s first chapter compares risk functions graphically and observes that when the curves cross “neither function is superior to the other” (p. 6), but never names the surviving property—finding uniform dominance rare, the book moves straight to the minimax criterion (Def. 2, p. 7). Among the probability texts, only DeGroot and Schervish define it, in the estimation chapter: an estimator is inadmissible if another *dominates* it, admissible if none does (Def. 7.9.1, p. 458)—and they immediately put the definition to work, using the Rao–Blackwell theorem (Thm. 7.9.1, p. 457) to manufacture dominating estima-

tors, and later recording Stein's startling result that the usual variance estimator is itself inadmissible (§8.7, pp. 510–511). Admissibility, in other words, is not a formality: reasonable-looking methods do fail it. ■

Problem 14

What is the basic principle that we adopt when comparing decision rules (statistical methods)? Your answer should please include the concept of risk and implicitly, but not explicitly, include the concept of admissibility.

SOLUTION:

The principle: judge a decision rule by its risk function, and try to minimize the risk (notes, Principle 1.5.12—“THIS IS A PRINCIPLE, NOT A RESULT”). Since a risk is an entire curve over Θ rather than a number, the operative form of the comparison is pointwise:

prefer δ to δ' when $R_\delta(\theta) \leq R_{\delta'}(\theta)$ for every θ , with strict inequality for some θ ,

and a rule that loses such a comparison to some competitor should be discarded—whatever the truth turns out to be, one would not regret the exchange. HPS states the same protocol on its sixth page: the comparison of decision functions “is most easily made by means of their graphs”; one function is better than another when “its risk function value is less than that of d_2 for each value of θ ,” since “our objective is to minimize the risk function” (p. 6).

The principle’s honesty lies in what it declines to promise. Risk curves cross—the rule that ignores the data and always reports 0 is unbeatable when the true difference is 0 and dreadful elsewhere—so “minimize the risk” cannot be a theorem-backed instruction; two rules need not be comparable at all, each better in its own region of Θ (HPS, p. 6: when the graphs cross, “neither function is superior to the other”). The principle therefore disciplines the search without completing it: it says what to throw

away, not which survivor to keep. Completing the choice requires either weighting the parameter space (the Bayesian route the assignment brackets) or guarding the worst case (HPS's minimax, Def. 2, p. 7). The property a rule earns by surviving every such comparison is the subject of Problem 13; the principle here uses only the comparison itself. ■

Risk Computations (Problems 15–16)

Suppose that Y has a binomial distribution with parameters n and p , n specified, but p unspecified. Suppose our loss function is $L(p, a) = (p - a)^2$. Consider two decision rules $d_0(Y) = Y/n$ and $d_1(Y) = Y/(n + c)$, for some positive constant c .

Problem 15

What are the risk functions for these two decision rules?

SOLUTION:

Under squared-error loss the risk is the mean squared error, so each risk function is a variance plus a squared bias (Problem 12); each computation therefore needs only the binomial moments $E^p\{Y\} = np$ and $\text{Var}^p(Y) = np(1 - p)$ and the scaling rules for expectation and variance. For $d_0(Y) = Y/n$:

$$E^p\{d_0(Y)\} = \frac{1}{n} E^p\{Y\} = \frac{np}{n} = p \quad (\text{scaling; binomial mean})$$

$$b_0(p) = p - p = 0 \quad (\text{unbiased})$$

$$\text{Var}^p(d_0(Y)) = \frac{1}{n^2} \text{Var}^p(Y) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n} \quad (\text{scaling; binomial variance})$$

$$R_{d_0}(p) = \text{Var}^p(d_0(Y)) + b_0(p)^2 = \boxed{\frac{p(1-p)}{n}} \quad (\text{bias-variance decomposition})$$

For $d_1(Y) = Y/(n + c)$:

$$E^p\{d_1(Y)\} = \frac{1}{n+c} E^p\{Y\} = \frac{np}{n+c} \quad (\text{scaling; binomial mean})$$

$$b_1(p) = \frac{np}{n+c} - p = -\frac{cp}{n+c} \quad (\text{definition of bias})$$

$$\text{Var}^p(d_1(Y)) = \frac{1}{(n+c)^2} \text{Var}^p(Y) = \frac{np(1-p)}{(n+c)^2} \quad (\text{scaling; binomial variance})$$

$$R_{d_1}(p) = \frac{np(1-p)}{(n+c)^2} + \frac{c^2 p^2}{(n+c)^2} = \boxed{\frac{np(1-p) + c^2 p^2}{(n+c)^2}} \quad (\text{bias-variance decomposition})$$

The ingredients are as standard as ingredients get—the binomial mean and variance are boxed in Ross (p. 132), derived by indicators in Blitzstein and Hwang (pp. 156, 174) and Pitman (pp. 169, 195), and recorded by DeGroot and Schervish (pp. 221, 231)—and the assembly is Problem 12’s decomposition, with d_0 landing in the unbiased special case: its risk *is* its variance, the situation HPS takes as the basis for comparing unbiased estimators (p. 15). The rule d_1 inflates the denominator, pulling the observed rate toward 0; it pays a bias of order $1/n$ for a variance smaller by a factor $n^2/(n+c)^2$. (The lecture’s shaded estimator $(1+X)/(2+n)$ makes the same wager anchored at $\frac{1}{2}$ rather than 0; notes, Example 1.8.3.)

Whether the wager pays depends on where p is—and, just as much, on how large n is. Setting $R_{d_1}(p) \leq R_{d_0}(p)$ and simplifying, the condition is

$$\frac{p}{1-p} \leq \frac{2}{c} + \frac{1}{n}, \quad \text{equivalently} \quad p \leq p^*(n) = \frac{2n+c}{(c+2)n+c},$$

so each rule wins on one side of a crossing point and loses on the other: neither dominates at *any* sample size, and the comparison principle of Problem 14 leaves both on the table no matter how large the trial. What changes with n is the terms of the trade, in two ways (Figure 3). The crossing point falls from $p^*(1) = \frac{2+c}{2+2c}$ toward its limit $2/(2+c)$ —for $c = 2$, from $\frac{2}{3}$ down toward $\frac{1}{2}$ —so the region where shrinkage wins settles rather than disappears. And the stakes collapse: both risks are of order $1/n$, but the squared bias that distinguishes d_1 is of order $1/n^2$, one order smaller, so the curves sit far apart at $n = 5$, nearly coincide at $n = 80$, and have ratio tending to one in the limit (Problem 16). Small samples make the choice of estimator a genuine

decision; large samples drown it. The trade-off can be kept alive asymptotically only by letting the shrinkage grow with the sample: taking $c \propto \sqrt{n}$ puts the squared bias and the variance on the same order at every n , which is precisely the design of the minimax estimator $(X + \sqrt{n}/2)/(n + \sqrt{n})$ that the course text graphs against X/n in its opening risk comparison (HPS, pp. 14–15, Figure 1).

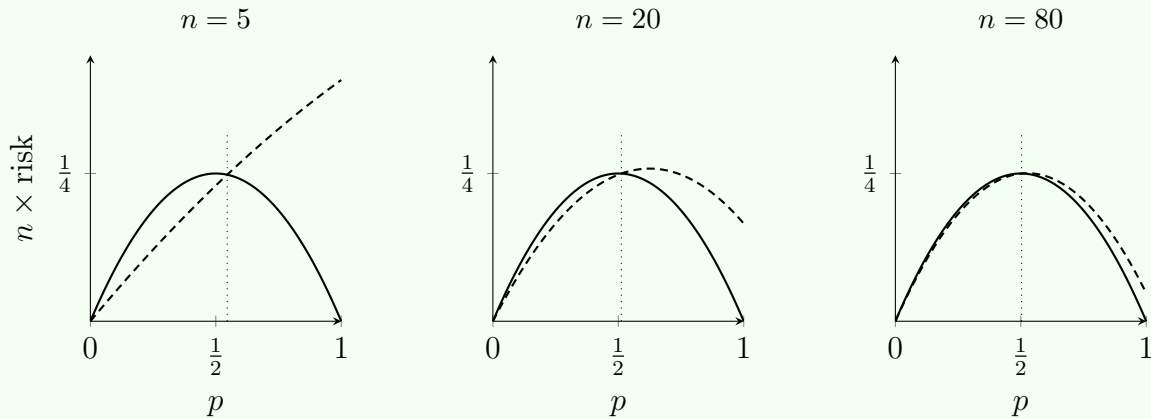


Figure 3. THE TRADE-OFF SHRINKS WITH THE SAMPLE. Normalized risks $nR(p)$ for $d_0 = Y/n$ (solid) and $d_1 = Y/(n + c)$ (dashed), $c = 2$; the normalization makes $nR_{d_0}(p) = p(1 - p)$ the same parabola in every panel, so the panels are directly comparable. The dotted line marks the crossing $p^*(n) = (2n + c)/((c + 2)n + c)$: shrinkage wins to its left, unbiasedness to its right. As n grows, the crossing drifts from 6/11 through 21/41 toward $\frac{1}{2}$, and the dashed curve collapses onto the solid one—at $n = 5$ the choice of rule matters; at $n = 80$ it barely does (Problem 16).



Problem 16

In the limit, as n goes to infinity, what is the ratio of these two risk functions?

SOLUTION:

Fix $p \in (0, 1)$ and let $c > 0$ stay fixed as n grows. Then

$$\begin{aligned}
 \frac{R_{d_1}(p)}{R_{d_0}(p)} &= \frac{np(1-p) + c^2p^2}{(n+c)^2} \cdot \frac{n}{p(1-p)} && \text{(form the ratio)} \\
 &= \frac{n^2 p(1-p) + n c^2 p^2}{(n+c)^2 p(1-p)} && \text{(multiply)} \\
 &= \frac{1 + \frac{c^2}{n} \cdot \frac{p}{1-p}}{\left(1 + \frac{c}{n}\right)^2} && \text{(divide through by } n^2\text{)} \\
 &\longrightarrow \boxed{1} \quad \text{as } n \rightarrow \infty. && \text{(both } 1/n \text{ corrections vanish)}
 \end{aligned}$$

The ratio tends to one: for every fixed p strictly between 0 and 1, the two decision rules are asymptotically equivalent. The bookkeeping explains why. Both risks are dominated by a variance of order $1/n$; d_1 's distinguishing feature, its bias, contributes $c^2 p^2 / (n+c)^2$, which is of order $1/n^2$ —one order smaller—so the shrinkage that visibly separates the rules in the small- n panels of Figure 3 is invisible at first order in the limit. The endpoints are the honest exceptions: at $p = 1$, R_{d_0} vanishes ($Y \equiv n$ makes d_0 exact) while $R_{d_1}(1) = c^2 / (n+c)^2 > 0$, so the ratio diverges rather than tending to one, and at $p = 0$ both risks are zero and the ratio is not defined. Neither endpoint disturbs the conclusion on the open interval, and the moral survives the caveat: asymptotics of this kind cannot distinguish estimators that differ only at order $1/n^2$, which is precisely why the finite-sample comparison of Problem 15—where the curves visibly cross—carries information the limit throws away. ■

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